Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Fisch, Allen J Ph.D.		
Project Title:	Development of a Haptic Interface for Remote Operations in Space, Medicine and Virtual Reality		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	avif@pinky.rutgers.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	732-445-4269
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City:	Piscataway	**State:**	NJ
Zip Code:	08854	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	09/01/2001	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To properly control remote robotic operations or to fully interact with a virtual reality world an interface is required that can accurately reflect visual, auditory and haptic (related to the sense of touch) information. Current technology can present sight and sound information to an operator, but forces cannot be completely reproduced. This project aims to develop a system that can reproduce compliance information, conveying an object's rigidity, using Electro-rheological fluids (ERFs). The viscosity of ERFs change in response to electrical stimuli. ERF actuators are being developed that will be incorporated into a lightweight system (e.g., a glove) which can be used for virtual or remote applications.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Haptic/Force Feedback system will facilitate a more intuitive interaction with robotic and Virtual Reality systems. Advanced robotic remote control, will allow, for example, human operators to remain in safety while deploying a robot in a difficult situation. The system will facilitate more immersive Virtual Reality simulations for training in medical, technical, and other fields. The Force Feedback system can also be used as an adjustable and programmable workout device for rehabilitation and other medical uses.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)